

A map you can draw in one walk

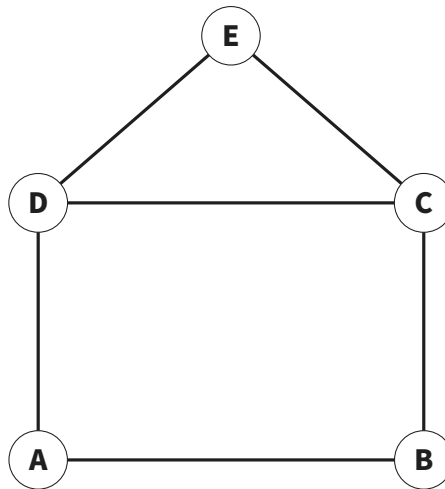
Name: _____

Date: _____

Travel every line exactly once without jumping. Visiting a dot more than once is allowed. You choose the start. Mark used lines with counters, or number them in walking order.

1. Find a route through the house.

Can you finish where you started? If not, find a walk with different starting and ending dots.



2. Pair an arrival with a departure.

At each dot, count the lines meeting there. This count is its **degree**. Circle the dots of odd degree.

Why must the start and finish be special?

Make a conjecture: How many odd dots can a one-walk map have? What changes if it must return to the start?

Try it with objects. Explain by pointing, talking, or drawing.

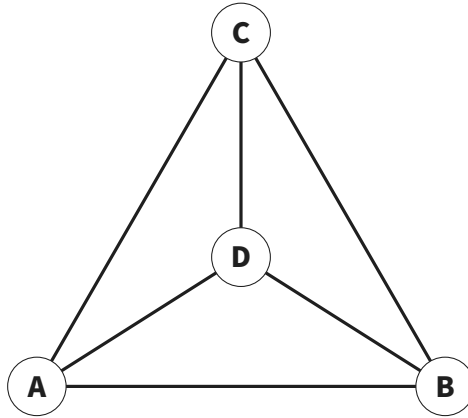
Repair an impossible map

Name: _____

Date: _____

3. One island inside three others.

Try to walk every line once. There are six lines and four dots. Does changing the start help?



An argument covering every possible start:

4. Add one bridge.

Draw a second bridge joining two dots already connected. Draw it as a curved line so both bridges are visible. Can the new map be walked using each bridge once? Choose the start carefully.

Which bridge did I add? My complete route:

5. Build a puzzle for a partner.

Draw a connected map with exactly two odd dots. Every bridge must join two different dots. Keep a complete one-walk solution hidden. Ask your partner where a route must start. If your map fails, revise it and explain what changed.

Try it with objects. Explain by pointing, talking, or drawing.